EMT Disconnected One-Point Building Blocks

Quark Loops, Gluon Loops, Stochastic Sources, and HDF5 Outputs

PyQUDA Measurement Notes

July 11, 2026

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1 Purpose and File Map

This note documents the hadron-independent one-point functions used to build disconnected contributions to proton and pion energy-momentum tensor (EMT) matrix elements. The main implementation is

`pyquda_measurement_utils/Disconnected_1pt_EMT_vibe_develop.py`.

The Perlmutter application entry points are in

`application/EMT_disconnected_1pt/perlmutter`.

The shorter command-oriented workflow guide is

`docs/EMT_disconnected_1pt/EMT_proton_disconnected_guide.md`.

The one-point workflows do not form hadron three-point functions by themselves. They measure vacuum loop building blocks. A disconnected hadron matrix element is formed later by correlating these one-point loops with hadron two-point functions on the same gauge ensemble.

2 Conventions

We work in Euclidean spacetime with four directions. Mathematical formulas use $\mu, \nu, \rho = 1, 2, 3, 4$, while Python loops use zero-based indices $\mu, \nu, \rho = 0, 1, 2, 3$. The temporal direction is the fourth Euclidean direction. PyQUDA stores local lattice fields in even-odd layout; the displayed formulas suppress this storage detail.

For a spatial momentum transfer q, the code constructs spatial phases with `MomentumPhase.getPhases`. In this note the phase is written as

$$\Phi_q(x) = e^{i\mathbf{q}\cdot\mathbf{x}}, \tag{1}$$

where only the spatial coordinates are Fourier transformed and the time coordinate t is kept explicit.

The spatial volume normalization used in the averaged HDF5 datasets is

$$V_3 = L_x L_y L_z. \tag{2}$$

The raw per-source quark datasets are stored before this V_3 normalization. The averaged quark and gluon datasets are volume-normalized.

3 Quark One-Point Workflow

The quark workflow is usually the first part to study. It contains the stochastic trace estimator, the hierarchical-probing source bookkeeping, the χ-type diagnostics, and the flowed quark EMT loop used in disconnected analysis.

3.1 Target Trace

A disconnected quark loop has the schematic form

$$L^q_\Gamma(q, t; t_f) = \sum_{\mathbf{x}} \Phi_q(x)\ \mathrm{Tr}_{sc}\big[\Gamma[U(t_f)](x)\, D^{-1}[U](x, x)\big]\,, \tag{3}$$

where D^{-1} is the quark propagator, Γ is the local operator kernel, and Tr_{sc} denotes spin-color trace. Directly computing all diagonal entries of $D^{-1}(x, x)$ is too expensive, so the trace is estimated with stochastic sources.

3.2 Stochastic Trace Estimator

Let ξ_r be a stochastic source vector whose spin-color-site components satisfy

$$\langle \xi_r \rangle_{\text{noise}} = 0, \qquad \left\langle \xi_r(i)\, \xi_r^\dagger(j) \right\rangle_{\text{noise}} = \delta_{ij}. \tag{4}$$

Here i, j are combined site, spin, and color indices. The code obtains

$$\eta_r = D^{-1}\xi_r \tag{5}$$

from the Dirac inverter. For any operator matrix A,

$$\left\langle \xi_r^\dagger A \eta_r \right\rangle_{\text{noise}} = \left\langle \xi_r^\dagger A D^{-1} \xi_r \right\rangle_{\text{noise}} \tag{6}$$

$$= \sum_{i,j,k} A_{ij}\,(D^{-1})_{jk} \left\langle \xi_{r,i}^\dagger \xi_{r,k} \right\rangle_{\text{noise}} \tag{7}$$

$$= \sum_{i,j} A_{ij}\,(D^{-1})_{ji} = \mathrm{Tr}\big(AD^{-1}\big). \tag{8}$$

This is the Hutchinson trace-estimator logic used by the quark one-point workflow.

3.3 Z_n Noise

The ordinary stochastic scheme is selected with

$$\texttt{noise\_scheme = zn}. \tag{9}$$

Each component of ξ is drawn independently from

$$\xi(i) \in \left\{ \exp\!\left(\frac{2\pi i r}{n}\right) \,\middle|\, r = 0, 1, \ldots, n-1 \right\}, \tag{10}$$

where $n =$ `n_zn`. The EMT quark one-point workflow now defaults to full-volume $\mathbb{Z}_4$ noise. Its counter-based generator hashes the global four-dimensional coordinate, spin, color, configuration number, base-noise index, and optional stream salt. Consequently the global source is reproducible and independent of the chosen MPI decomposition.

This counter construction fixes a serious failure mode of rank-local backend random-number generators. If every MPI rank resets an ordinary generator to the same state s, ranks with the same local lattice shape draw the same array. For a rank-local index ℓ,

$$\xi_r(\ell) = F(s, \ell) = \xi_{r'}(\ell), \qquad r \neq r'. \tag{11}$$

The corresponding distinct global sites $x_r + \ell$ and $x_{r'} + \ell$ then have unit rather than vanishing noise covariance,

$$\mathbb{E}_\xi[\xi^*(x_r + \ell)\xi(x_{r'} + \ell)] = 1, \qquad r \neq r', \tag{12}$$

so the required identity in Eq. (4) is not sampled. This creates rank-periodic off-diagonal terms, can bias the stochastic trace, and changes the global source when the MPI decomposition changes. Mixing the rank into the seed removes exact repetition only for one fixed process layout; it still makes the source depend on the rank assignment. The production counter instead uses physical global coordinates and semantic keys, so the same configuration, base index, and stream reconstruct exactly the same global field for every MPI geometry. Legacy backend-RNG loop data are therefore not combined with counter-noise production data.

3.4 Quark EMT Operator Implemented in Code

The quark EMT kernel in the one-point code is the local bilinear with a covariant derivative acting on the solution vector η. The implemented per-source, pre-volume-normalized object is

$$L_{\nu\mu}^{q,(r)}(q,t;t_f) = -\frac{1}{2}\sum_{\mathbf{x}} \Phi_q(x)\, \xi_r^\dagger(x;t_f)\, \gamma_\nu \left[D_{+\mu} - D_{-\mu}\right] \eta_r(x;t_f). \tag{13}$$

This formula maps directly to `_get_Tmunu_symmetrized_P_Breit_slice`. In code:

```
tmp = U_f.pure_gauge.covDev(eta, mu) - U_f.pure_gauge.covDev(eta, mu + 4)
Y = contract("ab,...bc->...ac", D_gammas_local[nu], tmp.data)
complex_field = contract("...sc,...sc->...", xi.data.conj(), Y)
Tmunu[nu, mu] += -0.5 * impose_P_Breit_slice(...)
```

The first tensor index is ν, because the gamma matrix γ_ν is applied after choosing the derivative direction μ. This is why the code fills `Tmunu[nu, mu]` before symmetrization.

3.5 Symmetrization

The EMT tensor is symmetrized after all unsymmetrized components have been accumulated:

$$L_{\mu\nu}^q \leftarrow \frac{1}{2}\left(L_{\mu\nu}^q + L_{\nu\mu}^q\right). \tag{14}$$

The code then sets both entries equal. The HDF5 writer stores only the upper triangle $\mu \le \nu$, with dataset names `T11`, `T12`, ..., `T44`. The lower triangle should be reconstructed by symmetry if needed.

3.6 Noise Average and Volume Normalization

For N_{eff} effective sources, the averaged quark one-point loop is

$$L_{\mu\nu,\text{avg}}^q(q,t;t_f) = \frac{1}{V_3}\frac{1}{N_{\text{eff}}}\sum_{r=1}^{N_{\text{eff}}} L_{\mu\nu}^{q,(r)}(q,t;t_f). \tag{15}$$

The code stores both the raw per-source data and the averaged volume-normalized data:

$$\texttt{raw/Tmunu\_pervec} \leftrightarrow L_{\mu\nu}^{q,(r)}(q,t;t_f), \tag{16}$$

$$\texttt{avg/Tmunu/Tij} \leftrightarrow L_{ij,\text{avg}}^q(q,t;t_f). \tag{17}$$

3.7 χ Diagnostics

The quark workflow also writes two scalar diagnostics:

$$\chi_0^{(r)}(q,t;t_f) = \sum_{\mathbf{x}} \Phi_q(x)\, \xi_r^\dagger(x;t_f)\, \eta_r(x;t_f), \tag{18}$$

$$\chi_1^{(r)}(q,t;t_f) = \sum_{\mathbf{x}} \Phi_q(x)\, \xi_r^\dagger(x;t_f)\, \xi_r(x;t_f). \tag{19}$$

χ_0 is the stochastic scalar trace estimate. χ_1 checks the noise normalization. In code these are the two entries of `CHI`:

```
dot_xi_eta = contract("etzyxbc,etzyxbc->etzyx", xi.data.conj(), eta.data)
CHI[0] = impose_P_Breit_slice(U_f, dot_xi_eta, phases_3pt)

dot_xi_xi = contract("etzyxbc,etzyxbc->etzyx", xi.data.conj(), xi.data)
CHI[1] = impose_P_Breit_slice(U_f, dot_xi_xi, phases_3pt)
```

The averaged HDF5 dataset `avg/CHI` is divided by V_3, just like the averaged quark EMT datasets.

3.8 Ringed-Fermion Kinetic Normalization

The raw `CHI` datasets are diagnostics, not the ringed-fermion normalization itself. The kinetic observable is already contained in the zero-momentum diagonal quark EMT components. For each effective stochastic source r, the implementation forms

$$K_r(t_f,\tau) = -\frac{2}{V_3}\sum_{\mu=1}^{4} L^q_{\mu\mu,r}(q=0,\tau;t_f) = \frac{1}{V_3}\sum_{\mu,\mathbf{x}} \xi^\dagger(x;t_f)\gamma_\mu\left[D_{+\mu} - D_{-\mu}\right]\eta(x;t_f). \tag{20}$$

No new inversion, fermion-flow update, covariant derivative, or MPI gather is needed: Eq. (20) is evaluated directly from the already-computed `raw/Tmunu_pervec`. The momentum list must contain exactly one $q = 0$ entry when companion output is enabled.

Every repository quark-1pt entry point writes a source-matched companion file

```
FlowedQuarkRinged/<lat>.FlowedQuarkRinged.<cfg>.<ama>.<sm>.h5
```

containing `raw/kinetic_pervec`, source bookkeeping, `avg/kinetic_spacetime`, and `flow_times`. It deliberately omits the two `avg/Z_ring` datasets: a factor computed from a single configuration is not the final ensemble normalization.

For configuration c, the stored spacetime average is

$$K_c(t_f) = \frac{1}{N_{\mathrm{eff}}N_t}\sum_{r,\tau} K_{r,c}(t_f,\tau). \tag{21}$$

The analysis must first form the gauge-ensemble mean

$$K_{\mathrm{code}}(t_f) = \langle K_c(t_f)\rangle_{\mathrm{cfg}}\,, \tag{22}$$

and only then, for a single-flavor trace in fundamental $SU(3)$, evaluate

$$Z^{(f)}_{\mathrm{ring}}(t_f) = \frac{-2N_c}{(4\pi)^2 t_f^2 K^{(f)}_{\mathrm{code}}(t_f)}. \tag{23}$$

In particular, $\langle 1/K_c\rangle_{\mathrm{cfg}} \neq 1/\langle K_c\rangle_{\mathrm{cfg}}$ in general, so configuration-local inverse factors must not be averaged. If instead the kinetic expectation value has already been summed or averaged over N_f degenerate flavors, the numerator and denominator must use the same flavor convention. A quark EMT bilinear made from ringed fields receives one factor Z_{ring}, because the square roots from χ and $\bar{\chi}$ multiply to this single bilinear factor. The $t_f = 0$ output step is kept as an unflowed diagnostic and must be excluded from Eq. (23). The standalone `application/flowed_quark_ringed_norm` workflow remains available for dedicated high-statistics and spin–color-diluted calculations. It uses the same base/HP-part validator and completion markers as EMT; no separate text sample log is used.

3.9 Gradient-Flow Schedule for Quarks

The quark workflow measures before flowing. Therefore step 0 is the unflowed observable. If $\epsilon = \texttt{flow\_epsilon}$, then

$$t_f(n) = n\,\epsilon, \qquad n = 0,\ldots,\texttt{flow\_steps}. \tag{24}$$

The first flow interval is internally split into ten smaller updates:

```
for step in range(Nsteps + 1):
    measure Tmunu and CHI
    if step == 0:
        flow [xi, eta] for 10 substeps of stepsize/10
    elif step < Nsteps:
        flow [xi, eta] for 1 step of stepsize
```

The stochastic source ξ and solution η are flowed together as one `MultiField`. This keeps both fields on the same flowed gauge background. After all flow steps are measured, the code extracts `raw/kinetic_pervec` algebraically from the stored $q = 0$ diagonal tensor entries and writes the kinetic companion on MPI rank zero.

3.10 Four-Dimensional Fermion Flow and the Time-Slice Projector

Let $K(t_f)$ denote the gauge-covariant fermion-flow kernel from flow time zero to t_f. Suppressing source labels, the fields measured by the code are

$$\xi_f = K(t_f)\xi, \qquad \eta_f = K(t_f)D^{-1}\xi. \tag{25}$$

The fermion flow is four dimensional: its covariant Laplacian couples all four Euclidean directions, including time. Its characteristic smoothing radius is approximately $\sqrt{8t_f}$. This radius is a diffusion scale, not a strict compact-support boundary.

Let P_τ project the measured output field onto the absolute Euclidean time slice τ, while leaving spatial, spin, and color indices unchanged. For an EMT kernel Γ on the flowed gauge background, one stochastic measurement can be written as

$$\begin{aligned}\widehat{L}(\tau, t_f) &= \xi_f^\dagger P_\tau \Gamma \eta_f \\ &= \xi^\dagger K^\dagger(t_f) P_\tau \Gamma K(t_f) D^{-1}\xi.\end{aligned} \tag{26}$$

Using the full-volume noise identity in Eq. (4),

$$\begin{aligned}\mathbb{E}_\xi[\widehat{L}(\tau, t_f)] &= \mathrm{Tr}\Big[K^\dagger(t_f) P_\tau \Gamma K(t_f) D^{-1}\Big] \\ &= \mathrm{Tr}\Big[P_\tau \Gamma K(t_f) D^{-1} K^\dagger(t_f)\Big].\end{aligned} \tag{27}$$

The projector P_τ means that the physical observable remains a spatial sum at a resolved insertion time; Eq. (27) is not a sum over insertion time. Nevertheless, the initial source must cover the full four-dimensional vector space because $K^\dagger P_\tau \Gamma K D^{-1}$ connects the measured time slice to source components at other times.

Equivalently, define

$$A_\tau = K^\dagger(t_f) P_\tau \Gamma K(t_f) D^{-1}. \tag{28}$$

For finite noise count, the off-diagonal stochastic contribution is

$$\widehat{L}(\tau, t_f) - L(\tau, t_f) = \sum_{i \neq j} \xi_i^* (A_\tau)_{ij} \xi_j. \tag{29}$$

The left index is constrained by the flowed time-slice operator, but the right index can originate from any Euclidean time. These terms vanish in the noise expectation; their finite-sample size is a variance question.

3.11 Why a Single Initial-Time Projector Is Not the Current Estimator

Suppose one replaces the full-volume source by only one time-diluted source $\xi_d = P_d\xi$. At nonzero flow time its expectation value is

$$\mathbb{E}_\xi\Big[\xi^\dagger P_d K^\dagger P_\tau \Gamma K D^{-1} P_d \xi\Big] = \mathrm{Tr}\Big[P_d K^\dagger P_\tau \Gamma K D^{-1} P_d\Big], \tag{30}$$

which is generally not Eq. (27). Four-dimensional fermion flow spreads the endpoint time slice over several initial-time components, so choosing only $d = \tau$ omits contributions.

Time dilution is not invalid in principle. For a complete orthogonal family

$$\sum_d P_d = \mathbf{1}, \qquad P_d P_{d'} = \delta_{dd'} P_d, \tag{31}$$

summing all diluted estimators reconstructs the full trace:

$$\sum_d \mathrm{Tr}[P_d A_\tau P_d] = \mathrm{Tr}\, A_\tau. \tag{32}$$

It requires all projectors and additional inversions. The present EMT workflow therefore uses one full-volume base source per stochastic solve and retains all absolute insertion times. In the unflowed limit $t_f = 0$, where $K(0) = \mathbf{1}$, a local loop at τ can be obtained from the matching exact-time projector; this special case does not justify dropping the other projectors for the finite-flow measurements stored in the same file.

4 Hierarchical Probing and Source Bookkeeping

The quark one-point workflow supports

$$\texttt{noise\_scheme} \in \{\texttt{zn}, \texttt{hierarchical\_probing}\}. \tag{33}$$

Hierarchical probing starts from a full-volume base noise vector ξ_b and multiplies it by deterministic site-only Rademacher vectors $h_k(x) = \pm 1$:

$$\xi_{b,k}(x) = h_k(x)\, \xi_b(x). \tag{34}$$

The probing vectors are designed to reduce near-diagonal contributions to the stochastic trace variance. In the current implementation the number of HP vectors must be a positive power of two:

$$\texttt{hp\_num\_vectors} = 1, 2, 4, 8, \ldots. \tag{35}$$

The effective number of inversions is

$$N_{\mathrm{eff}} = \begin{cases} \texttt{n\_vec}, & \texttt{noise\_scheme=zn}, \\ \texttt{n\_vec} \times \texttt{hp\_num\_vectors}, & \texttt{noise\_scheme=hierarchical\_probing}. \end{cases} \tag{36}$$

This value is stored as `effective_n_inversions`.

4.1 Ordering Choices

Four HP orderings are currently implemented:

Ordering	Meaning
`global_xyzt_gray_projected_to_evenodd`	Gray-code the full x, y, z, t site id.
`interleaved_xyzt_binary_projected_to_evenodd`	Interleave binary bits from x, y, z, t.
`interleaved_xyz_binary_projected_to_evenodd`	Interleave only spatial bits; the HP sign is time indeper
`spatial_xyz_then_t_gray_projected_to_evenodd`	Gray-code spatial sites first, then append time bits.

For the spatial ordering,

$$h_k(x, y, z, t) = h_k(x, y, z), \tag{37}$$

whereas a four-dimensional ordering has signs that also vary with t. Both cases multiply the same full-volume base noise and remain nonzero on every time slice; neither is time dilution. Spatial HP devotes all available probing bits to spatial locality, while isotropic 4D HP can suppress temporal off-diagonal noise at the cost of allocating fewer early bits to space. Their efficiency is observable dependent and should be compared at fixed inversion cost.

4.2 Bookkeeping Datasets

For each effective source, the output records:

Dataset	Meaning
raw/source_index	Effective source index after HP expansion.
raw/base_noise_index	Original base noise index.
raw/hp_index	HP vector index used for that base noise.

For example, with n_vec $= 1$ and hp_num_vectors $= 2$, the expected bookkeeping is

$$\texttt{source\_index} = [0, 1], \qquad \texttt{base\_noise\_index} = [0, 0], \qquad \texttt{hp\_index} = [0, 1]. \tag{38}$$

5 Quark HDF5 Layout

The production measurement writes validated base/HP shards and the explicit finalizer streams them into the canonical file. The main raw datasets are

$$\texttt{raw/Tmunu\_pervec} : [N_{\text{eff}}, 4, 4, N_q, N_{\text{steps}} + 1, N_t], \tag{39}$$

$$\texttt{raw/CHI\_pervec} : [N_{\text{eff}}, 2, N_q, N_{\text{steps}} + 1, N_t]. \tag{40}$$

The averaged datasets are

$$\texttt{avg/Tmunu/T11}, \texttt{T12}, \ldots, \texttt{T44} : [N_q, N_{\text{steps}} + 1, N_t], \tag{41}$$

$$\texttt{avg/CHI} : [2, N_q, N_{\text{steps}} + 1, N_t]. \tag{42}$$

Only $\mu \leq \nu$ tensor components are written in the averaged avg/Tmunu group.

The canonical loop file is source independent:

EMTc/<lat>.EMTc.<cfg>.<ama>.<sm>.h5.

It stores every absolute insertion time and is reused for all hadron two-point source times on the same configuration.

The source-matched kinetic companion is

FlowedQuarkRinged/<lat>.FlowedQuarkRinged.<cfg>.<ama>.<sm>.h5.

Its kinetic-only schema is

```
flow_times
raw/kinetic_pervec
raw/source_index
raw/base_noise_index
raw/hp_index
raw/spin_index
raw/color_index
avg/kinetic_spacetime
```

The spin and color indices are -1 because EMT uses full spin–color noise rather than point spin–color dilution. Attributes producer=emt_quark_1pt, content=kinetic_only, and ringed_factors_stored=Fal distinguish this subset from a full standalone output. The counter-noise configuration, stream, algorithm, HP ordering, Dirac parameters, gauge preprocessing, boundary condition, flow schedule, and exact EMT-to-kinetic relation are recorded alongside it.

5.1 Base-Oriented Production, Checkpoints, and Finalization

The production wrapper has a single shard output path; there is no monolithic production mode. A base noise and all of its hierarchical-probing vectors form the natural complete estimator unit,

$$L_b = \frac{1}{N_{\mathrm{HP}}} \sum_{h=0}^{N_{\mathrm{HP}}-1} L_{b,h}. \tag{43}$$

Inside one base the code writes fixed solve intervals, defaulting to 64 inversions. Thus HP256 normally produces four part files, while HP16 and plain $\mathbb{Z}_4$ produce one. A partial HP part is a recovery checkpoint and is not marked as a complete estimator.

Part names encode the base, part, and HP interval, for example

```
<canonical-stem>.base000003.part0001.hp0064-0127.h5
```

Only MPI rank zero writes HDF5. Each part is first written to a unique temporary path and then atomically renamed. Resume reopens existing parts and validates their physics metadata, shapes, and source indices; compatible parts are skipped, while conflicting or corrupt files cause an error. A base completion marker is published only after every expected part has been revalidated.

Measurement jobs do not update the canonical file. A separate serial finalizer requires complete coverage of bases $0, \ldots, N_{\mathrm{base}} - 1$, streams the raw datasets into a temporary canonical EMTc file, accumulates the averages without holding the complete source axis in memory, and derives the kinetic companion from the $q = 0$ diagonal tensor. The completed companion is published before the canonical EMTc file, so downstream analysis never sees a new loop without its matching kinetic data. Non-overlapping base ranges may be scheduled independently; overlapping jobs are unsupported.

One runtime detail is essential: evaluating a flowed contraction loads the flowed gauge into QUDA's global resident gauge state. Before inverting the next stochastic source, the code therefore reloads the original unflowed input gauge. The input Python gauge object is unchanged, but omitting this resident-state restore makes later inversions use the wrong gauge and can trigger precision mismatch failures.

The exact quark dataset and attribute names used by downstream scripts are:

```
raw/Tmunu_pervec
raw/CHI_pervec
raw/source_index
raw/base_noise_index
raw/hp_index
avg/Tmunu/T11
avg/Tmunu/T12
avg/Tmunu/T22
avg/Tmunu/T33
avg/Tmunu/T44
avg/CHI
noise_scheme
hp_num_vectors
effective_n_inversions
```

Important attributes include

Attribute	Meaning
`measurement`	`quark_1pt`.
`flow_type`, `flow_epsilon`, `flow_steps`, `flow_times`	Gradient-flow schedule.
`qext`, `pf`, `p_2pt`	Momentum labels used by downstream analysis.
`volume_norm`	V_3, the spatial-volume normalization.
`upper_triangle_only`	Whether only $\mu \leq \nu$ components are written.
`n_vec`, `n_base_noise`, `effective_n_inversions`	Stochastic source counts.
`noise_scheme`, `hp_num_vectors`, `hp_ordering`	Source-generation scheme.
`n_zn`, `config_num`, `noise_stream`	Counter-noise order, configuration key, and indepen
`noise_generator`, `noise_counter_order`	Fixed hash algorithm and the fields entering its cou
`operator_normalization`, `renormalization_applied`	Whether the saved quark loop has already been ren

6 Gluon One-Point Workflow

The gluon workflow is independent of stochastic quark sources. It constructs a flowed gauge-field strength and then measures a spatially Fourier-projected gluon EMT building block.

6.1 Clover Field Strength

For each independent plane (μ, ν), the code constructs the four oriented clover plaquettes

$$\mu \to \nu \to -\mu \to -\nu, \tag{44}$$

$$\nu \to -\mu \to -\nu \to \mu, \tag{45}$$

$$-\mu \to -\nu \to \mu \to \nu, \tag{46}$$

$$-\nu \to \mu \to \nu \to -\mu. \tag{47}$$

Let $Q_{\mu\nu}(x)$ be their sum. The anti-Hermitian clover field is

$$A_{\mu\nu}(x) = \frac{1}{8}\left[Q_{\mu\nu}(x) - Q^{\dagger}_{\mu\nu}(x)\right]. \tag{48}$$

The code projects this matrix to the traceless $\mathfrak{su}(3)$ algebra:

$$A^{\mathrm{tl}}_{\mu\nu}(x) = A_{\mu\nu}(x) - \frac{1}{N_c}\,\mathrm{Tr}_c[A_{\mu\nu}(x)]\,\mathbf{1}. \tag{49}$$

Finally, the Euclidean field-strength convention used by the implementation is

$$F_{\mu\nu}(x) = -i\,A^{\mathrm{tl}}_{\mu\nu}(x). \tag{50}$$

Only the six independent planes are computed, and the antisymmetric entries are filled with

$$F_{\nu\mu}(x) = -F_{\mu\nu}(x). \tag{51}$$

The traceless projection is a field-strength projection. It is not a traceless projection of the final EMT tensor.

6.2 Implemented Gluon EMT Loop

The implemented gluon one-point building block is

$$L^{g}_{\mu\nu}(q, t; t_f) = \frac{2}{V_3}\sum_{\mathbf{x}} \Phi_q(x) \sum_{\substack{\rho=1\\ \rho\neq\mu,\nu}}^{4} \mathrm{Tr}_c[F_{\mu\rho}(x; t_f)\, F_{\nu\rho}(x; t_f)]\,. \tag{52}$$

This maps directly to the code in `EMTDisconnectedGluon1pt.flowed_1pt`:

```
for mu in range(4):
    for nu in range(mu, 4):
        tmp = zeros(...)
        for rho in range(4):
            if rho == mu or rho == nu:
                continue
            tmp += contract("...ab,...ba->...", F[mu][rho], F[nu][rho])
        Tmunu_t[mu, nu, :, step, :] += 2.0 * slice_t
Tmunu_t /= Ns3
```

The factor 2 is explicit in the code. The division by V_3 is applied after all spatial Fourier sums have been accumulated.

6.3 Gradient-Flow Schedule for Gluons

As in the quark workflow, step 0 is measured before flow and therefore is the unflowed gauge observable. After each measurement, the gauge field is advanced by Wilson or Symanzik flow:

```
for step in range(Nsteps + 1):
    measure F and Tmunu
    if step == 0:
        flow gauge for 10 substeps of stepsize/10
    elif step < Nsteps:
        flow gauge for 1 step of stepsize
```

7 Gluon HDF5 Layout

The gluon writer is `save_emt_gluon_1pt_hdf5`. Unlike the quark file, there are no stochastic raw per-vector datasets. The output contains only volume-normalized upper-triangle tensor components:

$$\texttt{Tmunu/T11}, \texttt{T12}, \ldots, \texttt{T44} : [N_q, N_{\text{steps}} + 1, N_t]. \tag{53}$$

The canonical gluon loop is source independent,

`EMTg/<lat>.EMTg.<cfg>.<ama>.<sm>.h5`,

and is reused for all hadron source positions on the configuration. The shared quark and gluon wrappers use the same `EMT_1PT_FLOW_EPSILON`; its production default is (0.207936), so equal flow indices refer to equal flow times. Important attributes include

Attribute	Meaning
`measurement`	`gluon_1pt`.
`config_num`	Gauge configuration encoded by the canonical file.
`flow_type`, `flow_epsilon`, `flow_steps`, `flow_times`	Gradient-flow schedule.
`qext`, `pf`, `p_2pt`	Momentum labels used by downstream analysis.
`volume_norm`	V_3, the spatial-volume normalization.
`upper_triangle_only`	Whether only $\mu \leq \nu$ components are written.

8 Disconnected Combination with Proton Two-Point Functions

The quark and gluon one-point loops are hadron independent. They become a proton disconnected three-point building block only after being correlated with the proton two-point function on the gauge ensemble. For a proton two-point function $C_2^p(p_f, t)$, define

$$C_{3,\mu\nu}^{p,\mathrm{disc}}(p_f, p_i; t, \tau; t_f) = \langle C_2^p(p_f, t)\, L_{\mu\nu}(q, \tau; t_f)\rangle_{\mathrm{cfg}} - \langle C_2^p(p_f, t)\rangle_{\mathrm{cfg}} \langle L_{\mu\nu}(q, \tau; t_f)\rangle_{\mathrm{cfg}}\,, \tag{54}$$

with

$$q = p_f - p_i. \tag{55}$$

The subtraction term removes the vacuum expectation value of the one-point loop. This subtraction must be done at the ensemble-analysis level using the same configuration set as the proton two-point function.

A basic ratio is

$$R_{\mu\nu}^{p,\mathrm{disc}}(t, \tau; t_f) = \frac{C_{3,\mu\nu}^{p,\mathrm{disc}}(p_f, p_i; t, \tau; t_f)}{C_2^p(p_f, t)}. \tag{56}$$

Kinematic projectors, EMT mixing coefficients, and gradient-flow matching factors are applied later in the physics analysis.

9 End-to-End Proton Diagnostic Workflow

The application directory

`application/EMT_disconnected_1pt/perlmutter`

contains a minimal four-step diagnostic workflow:

1. measure the stochastic quark one-point loop;
2. measure the gluon one-point loop;
3. measure the proton two-point function C_2^p;
4. combine C_2^p and the one-point loops into disconnected diagnostic three-point objects.

For the quark loop, the builder reads `raw/Tmunu_pervec` and forms cumulative stochastic averages

$$\overline{L}_{\mu\nu}^{q,N}(q, t; t_f) = \frac{1}{V_3}\frac{1}{N}\sum_{r=1}^{N} L_{\mu\nu}^{q,(r)}(q, t; t_f), \qquad N = 1, \ldots, N_{\mathrm{eff}}. \tag{57}$$

This is the quantity used to check whether the stochastic estimator changes smoothly as more random sources are included.

For a single configuration, the builder can only form the unsubtracted proxy

$$C_{3,\mu\nu}^{\mathrm{unsub},N}(t_{\mathrm{sep}}, q, t; t_f) = C_2^p(t_{\mathrm{sep}})\, \overline{L}_{\mu\nu}^{q,N}(q, t; t_f). \tag{58}$$

This object is useful for checking shapes, source-count dependence, time alignment, and HDF5 plumbing. It is not the physical disconnected correlator, because the vacuum subtraction in Eq. (54) requires an ensemble average.

With multiple configurations, the builder applies the ensemble subtraction for each cumulative source count:

$$C_{3,\mu\nu}^{\mathrm{disc},N} = \left\langle C_2^p\, \overline{L}_{\mu\nu}^{q,N}\right\rangle_{\mathrm{cfg}} - \langle C_2^p\rangle_{\mathrm{cfg}} \left\langle \overline{L}_{\mu\nu}^{q,N}\right\rangle_{\mathrm{cfg}}\,, \tag{59}$$

$$R_{\mu\nu}^{\mathrm{disc},N} = \frac{C_{3,\mu\nu}^{\mathrm{disc},N}}{\langle C_2^p\rangle_{\mathrm{cfg}}}. \tag{60}$$

The gluon loop has no stochastic source axis, so the builder forms the same unsubtracted and vacuum-subtracted objects without a cumulative N index.

The diagnostic merger output contains:

```
C2
quark/source_count
quark/loop_cumulative
quark/C3_unsubtracted_cumulative
quark/C3_disc_cumulative     # only if Ncfg >= 2
quark/R_disc_cumulative      # only if Ncfg >= 2
gluon/loop
gluon/C3_unsubtracted
gluon/C3_disc                # only if Ncfg >= 2
gluon/R_disc                 # only if Ncfg >= 2
summary/quark_source_count
summary/quark_T44_q0_flow0_loop_norm
summary/quark_T44_q0_flow0_unsub_ratio_proxy
```

10 How to Read Smoke Tests

A single `S8T32` test gauge checks implementation conventions, not physics signals. Useful smoke-test checks are:

- the code runs on the intended backend;
- `flow_times` contains step 0 and the requested flow times;
- `upper_triangle_only=True`;
- quark raw datasets have the expected $[N_{\mathrm{eff}}, 4, 4, N_q, N_{\mathrm{steps}}+1, N_t]$ and $[N_{\mathrm{eff}}, 2, N_q, N_{\mathrm{steps}}+1, N_t]$ shapes;
- HP runs have consistent `raw/source_index`, `raw/base_noise_index`, and `raw/hp_index`;
- gluon datasets contain `Tmunu/T11` through `T44` but no stochastic `raw/Tmunu_pervec`.

11 Recommended Next Algorithmic Upgrade

The current variance-reduction options are ordinary Z_n noise and hierarchical probing. A planned future direction is multiplier-based graph coloring, motivated by

arXiv:2606.00394.

The motivation is that standard HP naturally tests 2^k source counts, whereas multiplier-based coloring can test intermediate coloring budgets. A future scheme should be added as a separate optional source scheme, with HDF5 metadata that clearly distinguishes it from `hierarchical_probing`.